

From Raw Echoes to a Focused SAR Image

A compact derivation of the Range Migration Algorithm
for a three-dimensional stripmap geometry

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Abstract

This note accompanies the Python notebook `SimulationSAR_RMA_3D.ipynb`. Its purpose is to make explicit both the inverse problem solved by synthetic-aperture radar and the calculations that connect a three-dimensional monostatic stripmap acquisition to a focused ground-range/azimuth image. The recorded data are the complex raw traces $s(t, u)$, concatenated over fast time t and aperture position u . The unknown image is a complex reflectivity map $\rho(x, y)$: after discretization, each image pixel represents the coefficient of a hypothetical point scatterer located at (x, y) . The Range Migration Algorithm provides a structured inversion of the corresponding forward operator. The derivation follows the notation and implementation choices of the notebook: exact propagation in three dimensions, reduction to a two-dimensional slant-coordinate problem, fast-time range filtering, Fourier transformation along the aperture, stationary-phase evaluation, spectral phase compensation, Stolt interpolation, inverse two-dimensional Fourier transformation, and complex remapping from slant range to ground range. The presentation is intentionally compact, but no essential computational step is omitted.

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1 Scope and Fourier conventions

A Synthetic-Aperture Radar does not acquire an image directly. It records a complex signal indexed by fast time t and by the antenna position u along the synthetic aperture. The image is recovered by inverting the propagation model coherently.

Throughout this note, the one-dimensional Fourier transform is defined by

$$X(\kappa) = \int_{-\infty}^{+\infty} x(\xi) e^{-j\kappa\xi} d\xi, \quad x(\xi) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\kappa) e^{+j\kappa\xi} d\kappa. \quad (1)$$

The same sign convention is used independently along the range and azimuth coordinates. Numerical FFT normalization factors depend on the transform convention and on the software library. Constant normalization factors do not affect target localization or focusing, and are therefore omitted here whenever they would obscure the geometry. They must, however, be tracked carefully when absolute amplitudes, powers, or radiometric quantities are interpreted.

We use the following distinction:

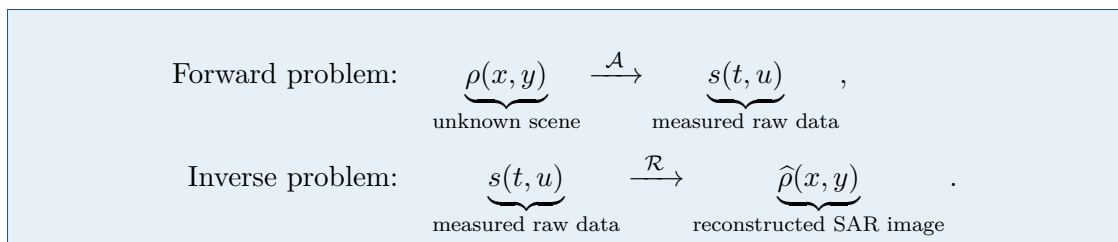
- f is the radio frequency and $\omega = 2\pi f$ the angular frequency;
- $k = \omega/c = 2\pi f/c$ is the one-way electromagnetic wavenumber;
- because the radar path is monostatic (same position for the emission and the reception), the propagation phase contains the two-way wavenumber $2k$;
- k_y is the spatial frequency conjugate to azimuth y ;
- k_q is the spatial frequency conjugate to the cross-track slant coordinate q .

2 The inverse problem and the meaning of the image

At each antenna position u_m , the radar records one complex fast-time trace $s(t, u_m)$. Sampling fast time at t_n and concatenating all traces gives the raw-data matrix

$$\mathbf{S}_{\text{raw}} = [s(t_n, u_m)] \in \mathbb{C}^{N_t \times N_u}. \quad (2)$$

This matrix is what is commonly called the *raw SAR signal*. Its horizontal coordinate is aperture position, its vertical coordinate is fast time, and its entries are complex in-phase/quadrature samples. It is not yet an image.



The unknown scene is modelled by a complex ground-reflectivity map $\rho(x, y)$, where x is ground range relative to the scene centre and y is azimuth. A useful way to interpret this map is to imagine an elementary point scatterer at every image pixel. After discretization,

$$\rho(x, y) \simeq \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \rho_{ij} \delta(x - x_i) \delta(y - y_j), \quad (3)$$

where the complex coefficient ρ_{ij} is the backscattered amplitude and phase assigned to pixel (x_i, y_j) . The sought SAR image is the two-dimensional arrangement of the estimated coefficients $\hat{\rho}_{ij}$, not the raw matrix $\mathbf{S}_{\text{raw}}$.

For a fixed acquisition geometry and transmitted waveform, propagation defines a linear forward operator $\mathcal{A}$:

$$\mathbf{s} = \mathcal{A}\boldsymbol{\rho} + \mathbf{n}, \quad \mathbf{s} = \text{vec}(\mathbf{S}_{\text{raw}}), \quad (4)$$

where $\boldsymbol{\rho}$ contains all unknown pixel coefficients and $\mathbf{n}$ represents measurement noise and modelling errors. SAR focusing is therefore an inverse problem:

$$\hat{\boldsymbol{\rho}} = \mathcal{R}\{\mathbf{s}\}, \quad \mathcal{R} \simeq \mathcal{A}^\dagger, \quad (5)$$

where $\mathcal{R}$ denotes the reconstruction algorithm and $\mathcal{A}^\dagger$ represents an idealized inverse or pseudoinverse. The RMA does not form or invert the enormous matrix $\mathcal{A}$ explicitly. It exploits the structure of the propagation phase so that the inversion can be implemented through filtering, Fourier transforms, phase compensation, and interpolation.

The reconstructed quantity is complex. A displayed amplitude image is usually $|\hat{\rho}(x, y)|$, while a power image is proportional to $|\hat{\rho}(x, y)|^2$; either may be shown on a logarithmic scale. This distinction matters: the display is derived from the estimated complex reflectivity map, whereas coherent processing uses the complex map itself.

3 Three-dimensional stripmap geometry

The radar moves along a straight line parallel to the Y axis at altitude H . Its position at aperture coordinate u is

$$\mathbf{r}_r(u) = (0, u, H). \quad (6)$$

The flat ground is the plane $Z = 0$. The scene centre is located at ground range X_c , and a point scatterer with relative coordinates (x_p, y_p) is located at

$$\mathbf{r}_p = (X_c + x_p, y_p, 0). \quad (7)$$

The exact one-way propagation distance is therefore

$$R_p(u) = \|\mathbf{r}_p - \mathbf{r}_r(u)\| = \sqrt{(X_c + x_p)^2 + (y_p - u)^2 + H^2}. \quad (8)$$

This expression controls both the propagation delay and the carrier phase.

Define the cross-track slant coordinate

$$q_p = \sqrt{(X_c + x_p)^2 + H^2}. \quad (9)$$

Then eq. (8) becomes

$$R_p(u) = \sqrt{q_p^2 + (u - y_p)^2}. \quad (10)$$

This is formally the standard two-dimensional range-azimuth geometry. The omega- k processing can therefore be performed in (q, y) , even though the original acquisition is three-dimensional.

The relation between absolute ground range X and cross-track slant coordinate q is

$$q(X) = \sqrt{X^2 + H^2}, \quad X(q) = \sqrt{q^2 - H^2}. \quad (11)$$

The reconstruction will first produce an image sampled uniformly in q , after which a complex interpolation is required to obtain a uniform ground-range axis X .

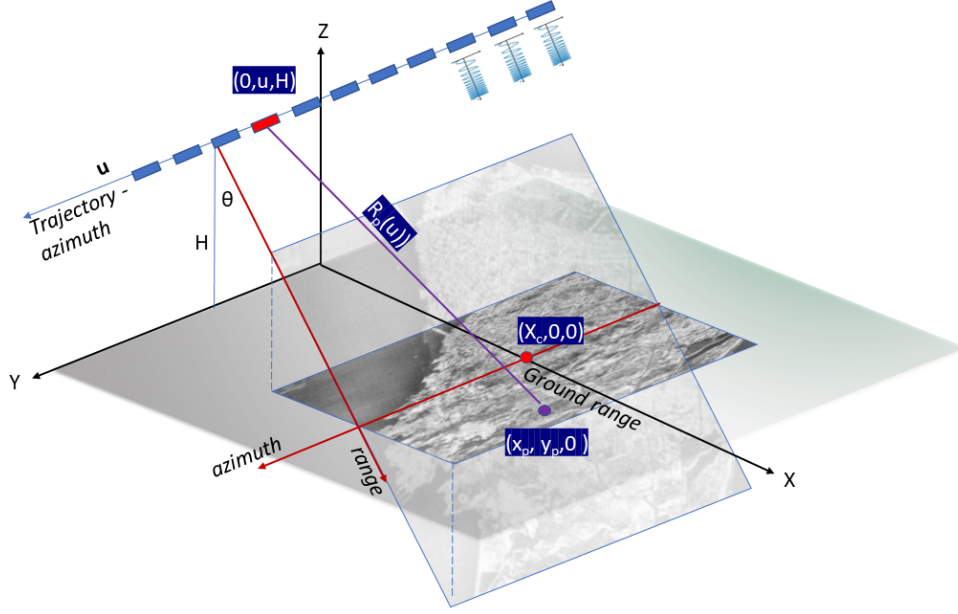


Figure 1: Three-dimensional stripmap acquisition geometry. The antenna positions are indexed by the aperture coordinate u at altitude H . The scene centre is located at $(X_c, 0, 0)$, and the instantaneous radar-to-target distance is $R_p(u)$. Ground range and azimuth define the two coordinates of the reflectivity image.

4 Waveform and nominal resolutions

The notebook uses an X-band linear frequency-modulated pulse.

What “baseband” means. The physical waveform transmitted by the antenna oscillates around the carrier frequency f_c . A real radio-frequency signal can be written as

$$e_{\text{RF}}(t) = \Re\{p(t) e^{j\omega_c t}\}, \quad \omega_c = 2\pi f_c, \quad (12)$$

where $p(t)$ is the slowly varying complex envelope. Quadrature demodulation multiplies the received RF signal by $e^{-j\omega_c t}$ and low-pass filters the result. The remaining in-phase and quadrature components form the complex baseband signal $I(t) + jQ(t)$. Thus, “baseband” does not mean that phase information has been removed: it means that the rapid carrier oscillation has been translated from f_c to zero frequency while its amplitude and phase modulation are retained. In the notebook, $s(t, u)$ denotes these complex baseband raw data before SAR focusing.

The complex baseband chirp is

$$p(t_d) = \exp\{j[-\pi B t_d + \alpha t_d^2]\} \mathbf{1}_{[0, T_p]}(t_d), \quad \alpha = \frac{\pi B}{T_p}. \quad (13)$$

The instantaneous baseband frequency follows from the derivative of the phase:

$$f_{\text{bb}}(t_d) = \frac{1}{2\pi} \frac{d}{dt_d} (-\pi B t_d + \alpha t_d^2) = -\frac{B}{2} + \frac{B}{T_p} t_d. \quad (14)$$

Thus the radio frequency sweeps from $f_c - B/2$ to $f_c + B/2$.

The nominal slant-range resolution is

$$\delta q \simeq \frac{c}{2B}. \quad (15)$$

At the scene centre, where the incidence angle measured from the vertical is

$$\theta_c = \arctan\left(\frac{X_c}{H}\right), \quad (16)$$

the corresponding ground-range resolution is approximately

$$\delta X \simeq \frac{\delta q}{\sin \theta_c} = \frac{c}{2B \sin \theta_c}. \quad (17)$$

Table 1: Principal parameters used in the notebook.

Quantity	Symbol	Value
Carrier frequency	f_c	9.6 GHz
Bandwidth	B	150 MHz
Pulse duration	T_p	5 μ s
Carrier wavelength	$\lambda_c = c/f_c$	3.125 cm
Altitude	H	1000 m
Scene-centre ground range	X_c	3000 m
Scene-centre slant range	R_c	3162.28 m
Central incidence angle	θ_c	71.57°
Nominal slant-range resolution	δq	1.00 m
Nominal ground-range resolution	δX	1.05 m
Target azimuth resolution	δy	1.00 m
Synthetic-aperture length	L_{sa}	49.41 m

The pulse duration controls the transmitted energy and the time–bandwidth product BT_p , but not the range resolution by itself. The azimuth resolution is governed separately by the angular diversity collected over the synthetic aperture.

5 Forward operator and raw complex signal

For a point scatterer p , the two-way propagation delay is

$$\tau_p(u) = \frac{2R_p(u)}{c}. \quad (18)$$

With $t_d = t - \tau_p(u)$, its contribution to the complex baseband data is the delayed chirp multiplied by the two-way carrier phase. For a set of point scatterers,

$$s(t, u) = \sum_p \rho_p \exp\left\{j \left[-\omega_c \tau_p(u) - \pi B t_d + \alpha t_d^2 \right] \right\} \mathbf{1}_{[0, T_p]}(t_d), \quad (19)$$

where ρ_p is the unknown complex reflectivity of the p -th point. Evaluating this expression at every sampled pair (t_n, u_m) fills the raw-data matrix in eq. (2). In other words, each column is one received trace, and the complete raw signal is obtained by concatenating the traces recorded along the aperture.

The first phase term can be written

$$-\omega_c \tau_p(u) = -\frac{4\pi f_c}{c} R_p(u) = -\frac{4\pi}{\lambda_c} R_p(u). \quad (20)$$

It is this rapidly varying phase that carries most of the azimuth information. A range change much smaller than one range-resolution cell can still produce many phase cycles because λ_c is only a few centimetres.

For a continuous ground-reflectivity map $\rho(x, y)$, the forward operator is

$$s(t, u) = \iint \rho(x, y) p\left(t - \frac{2R(x, y; u)}{c}\right) \exp\left[-j \frac{4\pi f_c}{c} R(x, y; u)\right] dx dy, \quad (21)$$

with

$$R(x, y; u) = \sqrt{(X_c + x)^2 + (u - y)^2 + H^2}. \quad (22)$$

This equation makes the inverse problem explicit: the data $s(t, u)$ are known, while the two-dimensional function $\rho(x, y)$ is unknown.

After the change of coordinate

$$q = \sqrt{(X_c + x)^2 + H^2}, \quad (23)$$

the geometry becomes $R(q, y; u) = \sqrt{q^2 + (u - y)^2}$. We denote by $\rho_q(q, y)$ the same scene expressed in the slant coordinate, with the coordinate Jacobian absorbed into the reflectivity-density convention. The forward model then becomes

$$s(t, u) = \iint \rho_q(q, y) p\left(t - \frac{2R(q, y; u)}{c}\right) \exp\left[-j\frac{4\pi f_c}{c}R(q, y; u)\right] dq dy. \quad (24)$$

For the discrete point-scatterer model used in the notebook, no density Jacobian is required: the coefficients ρ_p are simply transported from their ground coordinates to the corresponding slant coordinates.

A slowly varying amplitude factor, for example $1/R^2$, may be added. It is disabled in the baseline notebook to isolate the geometrical focusing mechanism.

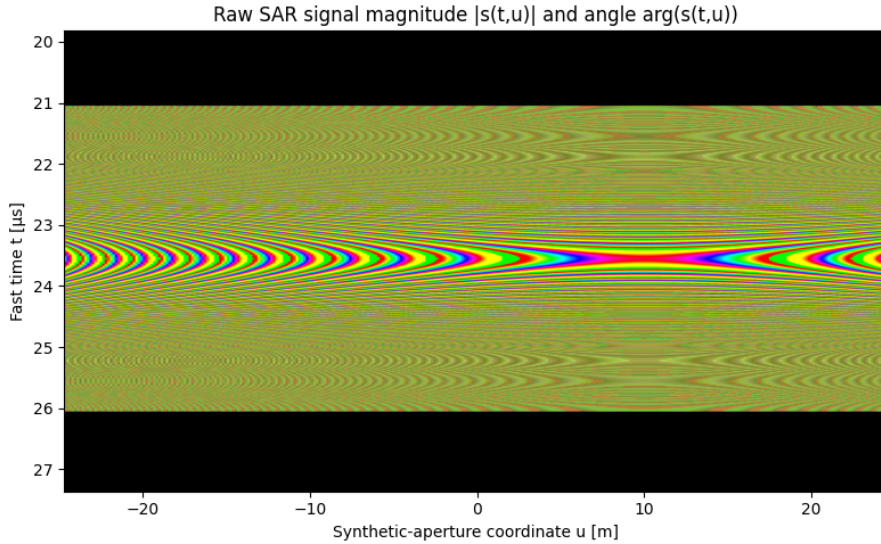


Figure 2: Magnitude and phase of the simulated raw-data matrix $s(t_n, u_m)$. Brightness represents amplitude and hue represents phase ; the useful azimuth information remains present in the complex phase.

6 Fast-time range filtering

Let $s_0(t)$ be a reference echo corresponding to a ground point at $X = X_c + x_{\text{ref}}$, zero azimuth, and zero-Doppler slant range

$$q_{\text{ref}} = \sqrt{(X_c + x_{\text{ref}})^2 + H^2}. \quad (25)$$

The notebook computes centred fast-time Fourier transforms

$$S_{\text{raw}}(f, u) = \mathcal{F}_t\{s(t, u)\}, \quad S_0(f) = \mathcal{F}_t\{s_0(t)\}. \quad (26)$$

Two filters are provided. The matched filter is

$$H_{\text{MF}}(f) = S_0^*(f), \quad (27)$$

whereas the normalized or “whitened” filter is

$$H_W(f) = \frac{S_0^*(f)}{|S_0(f)|^2 + \varepsilon \max_f |S_0(f)|^2}. \quad (28)$$

The range-filtered spectrum is

$$S_r(f, u) = S_{\text{raw}}(f, u)H(f). \quad (29)$$

The matched filter maximizes the output signal-to-noise ratio in additive white noise. The normalized filter behaves instead as a regularized deconvolution. It produces a particularly clean impulse response in a noiseless simulation, but it may amplify noise where $S_0(f)$ is weak.

After removal of the transmitted waveform and of the phase associated with the reference range, the useful part of the spectrum can be written, up to a slowly varying amplitude factor, as

$$S_r(k, u) \propto \iint \rho_q(q, y) \exp[-j2kR(q, y; u)] \exp[+j2kR_c] dq dy, \quad (30)$$

where $R_c = \sqrt{X_c^2 + H^2}$. This expression is the starting point of the omega- k derivation.

7 Fourier transform along the aperture

The aperture Fourier transform is

$$S_r(k, k_y) = \int_{-\infty}^{+\infty} S_r(k, u) e^{-jk_y u} du. \quad (31)$$

For one elementary scatterer at (q, y) , the relevant integral is

$$I(q, y; k, k_y) = \int_{-\infty}^{+\infty} \exp\left[-j2k\sqrt{q^2 + (u - y)^2}\right] \exp(-jk_y u) du. \quad (32)$$

This integral is oscillatory and has no elementary closed form in the representation useful for SAR processing. The Range Migration Algorithm uses its stationary-phase approximation.

Define the phase

$$\phi(u) = -2k\sqrt{q^2 + (u - y)^2} - k_y u. \quad (33)$$

Its derivative is

$$\phi'(u) = -2k \frac{u - y}{\sqrt{q^2 + (u - y)^2}} - k_y. \quad (34)$$

At the stationary point u_s , $\phi'(u_s) = 0$, hence

$$\frac{u_s - y}{\sqrt{q^2 + (u_s - y)^2}} = -\frac{k_y}{2k}. \quad (35)$$

Introduce

$$k_q = \sqrt{4k^2 - k_y^2}. \quad (36)$$

The propagating domain requires $|k_y| < 2k$. Solving eq. (35) gives

$$u_s - y = -\frac{k_y}{k_q} q, \quad R_s = \sqrt{q^2 + (u_s - y)^2} = \frac{2k}{k_q} q. \quad (37)$$

Substitution into the phase yields

$$\phi(u_s) = -k_q q - k_y y. \quad (38)$$

The second derivative is

$$\phi''(u) = -2k \frac{q^2}{[q^2 + (u - y)^2]^{3/2}}, \quad (39)$$

and at the stationary point

$$\phi''(u_s) = -\frac{k_q^3}{4k^2q} < 0. \quad (40)$$

The stationary-phase theorem then gives

$$I(q, y; k, k_y) \simeq \sqrt{\frac{8\pi k^2 q}{k_q^3}} \exp\left(-j\frac{\pi}{4}\right) \exp[-j(k_q q + k_y y)]. \quad (41)$$

The square-root factor varies more slowly than the phase. The baseline notebook neglects it for geometrical focusing. Retaining or compensating this factor becomes relevant for accurate radiometry.

8 Two-dimensional phase compensation

Using eq. (41) in eq. (30), and neglecting the slowly varying amplitude, the aperture spectrum takes the form

$$S_r(k, k_y) \propto \iint \rho_q(q, y) \exp[-j(k_q q + k_y y)] \exp(+j2kR_c) \exp\left(-j\frac{\pi}{4}\right) dq dy. \quad (42)$$

The notebook applies the phase filter

$$H_{2D}(k, k_y) = \exp\left\{j\left[k_q R_c + \frac{\pi}{4} - 2kR_c\right]\right\}. \quad (43)$$

The compensated spectrum is therefore

$$S_c(k, k_y) \propto \iint \rho_q(q, y) \exp[-jk_q(q - R_c) - jk_y y] dq dy. \quad (44)$$

This is already a two-dimensional Fourier representation of the scene in the variables (k_q, k_y) . The remaining difficulty is that the samples are not uniformly spaced in k_q .

9 Stolt interpolation

The acquisition is sampled uniformly in radio frequency f , hence uniformly in k , and uniformly in aperture position u , hence uniformly in k_y after the aperture FFT. However,

$$k_q(k, k_y) = \sqrt{4k^2 - k_y^2} \quad (45)$$

is nonlinear. For each fixed k_y , uniform k samples map to nonuniform k_q samples.

A two-dimensional inverse FFT requires a Cartesian grid (K_q, k_y) with constant spacing in both directions. Stolt interpolation estimates the compensated spectrum on this regular grid:

$$F(K_q, k_y) = \text{Interp}_{k_q \rightarrow K_q} \{S_c[k_q(k, k_y), k_y]\}. \quad (46)$$

The notebook performs this interpolation independently for each k_y and separately on the real and imaginary parts. Linear interpolation is used by default because it is robust for highly oscillatory complex spectra.

A change of variables in a continuous integral introduces

$$\frac{\partial k}{\partial k_q} = \frac{k_q}{4k}. \quad (47)$$

The notebook provides this factor as an optional Stolt Jacobian. It is disabled in the baseline experiment because the primary objective is localization and point-spread-function shape rather than calibrated radiometry.

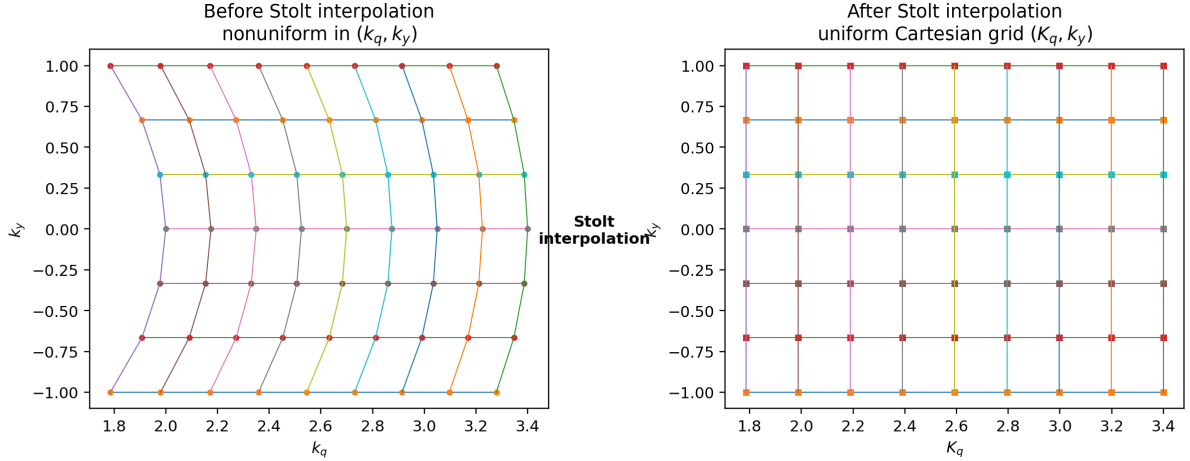


Figure 3: Principle of Stolt interpolation. The original samples are uniform in (k, k_y) but occupy a warped, nonuniform grid after the nonlinear mapping to k_q . They are interpolated onto a regular Cartesian grid (K_q, k_y) before the inverse FFT.

10 Inverse transform in slant coordinates

After interpolation, the spectrum is sampled on a Cartesian grid. The complex image in slant coordinates is obtained by

$$\hat{\rho}_q(R_c + q', y) = \mathcal{F}_{K_q, k_y}^{-1} \{F(K_q, k_y)\}, \quad q' = q - R_c. \quad (48)$$

The absolute slant coordinate is then $q = R_c + q'$.

The discrete spatial sampling intervals follow from the spectral spacings:

$$\Delta q = \frac{2\pi}{N_q \Delta K_q}, \quad \Delta y = \frac{2\pi}{N_y \Delta k_y}. \quad (49)$$

Zero-padding the spectrum increases N_q and therefore decreases the displayed sampling interval Δq , but it does not enlarge the measured spectral support. Consequently, it interpolates the PSF without improving the physical resolution.

FFT centring must be handled consistently. In the notebook, centred FFT and inverse FFT utilities combine `fftshift` and `ifftshift`; the reconstructed complex reflectivity array is indexed internally as $\hat{\rho}_q[q, y]$ and is transposed for conventional image display as $\hat{\rho}_q[y, q]$.

11 Complex remapping to ground range

The image delivered by the omega- k inversion is uniform in q . A uniform ground-range grid X_i is selected and mapped to

$$q_i = \sqrt{X_i^2 + H^2}. \quad (50)$$

For every azimuth row, the notebook interpolates the real and imaginary parts separately:

$$\hat{\rho}_g(X_i, y) = \text{Interp}\{\Re \hat{\rho}_q(q, y)\}_{q=q_i} + j \text{Interp}\{\Im \hat{\rho}_q(q, y)\}_{q=q_i}. \quad (51)$$

Only after this operation is the displayed amplitude formed. The resulting array $\hat{\rho}_g(X, y)$ is the complex SAR image: one estimated reflectivity coefficient for every ground-range/azimuth pixel.

$$I_{\text{display}}(X, y) = |\hat{\rho}_g(X, y)|. \quad (52)$$

Interpolating the magnitude instead would irreversibly discard phase information and would make the result unsuitable for coherent operations such as interferometry or coherent combination of channels.

12 Point-target validation

The single-scatterer experiment is the most sensitive diagnostic of the processor. The notebook reconstructs one point target at the scene centre. The maximum appears at the expected ground-range and azimuth coordinates to within a small fraction of a resolution cell.

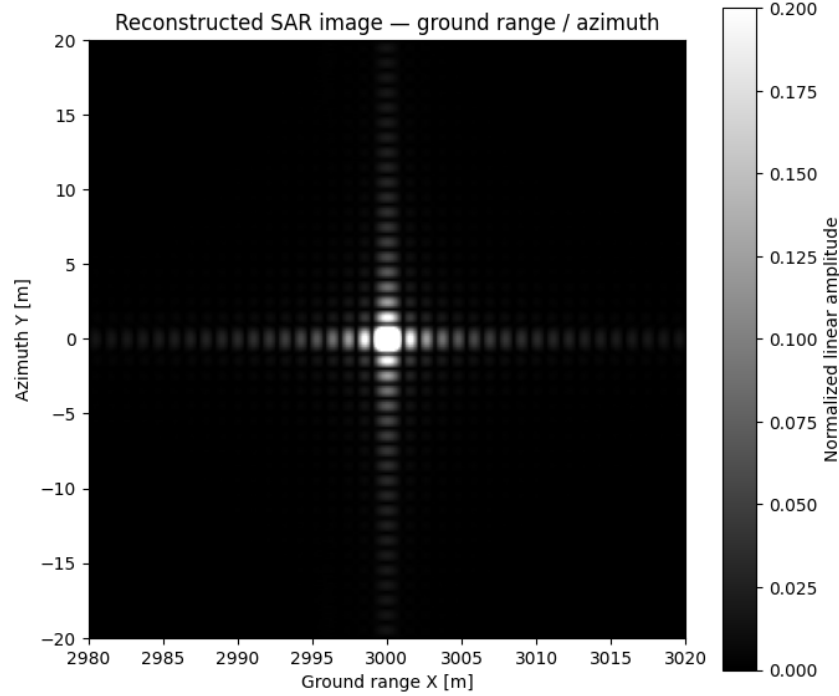


Figure 4: Focused response of one point scatterer in ground-range/azimuth coordinates. The finite bandwidth and finite aperture produce a two-dimensional point-spread function rather than an ideal pixel.

The finite bandwidth produces the range main lobe and sidelobes; the finite synthetic aperture produces the azimuth main lobe and sidelobes. With the notebook parameters, the measured full widths at half maximum are close to the nominal one-metre resolutions.

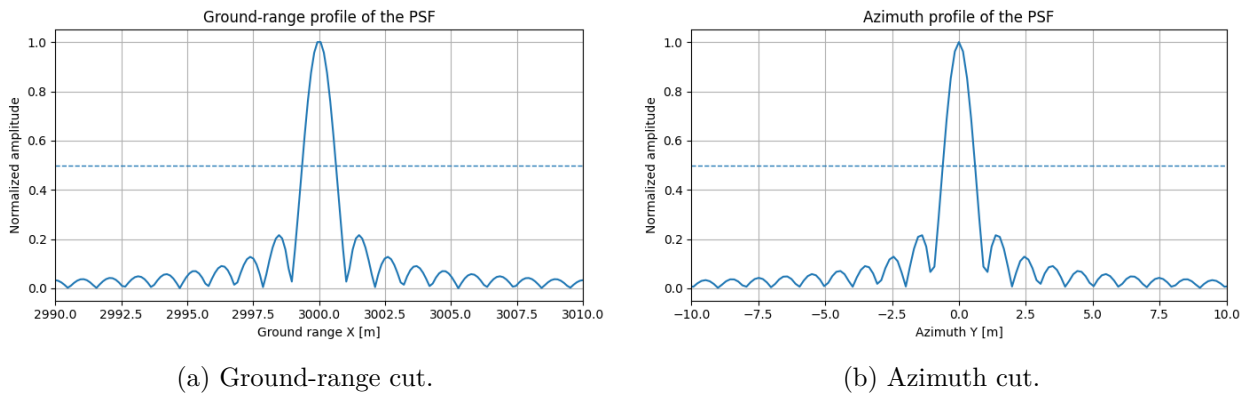


Figure 5: Point-spread-function profiles through the reconstructed maximum. The dashed line marks half amplitude.

The nearly horizontal raw envelope in fig. 2 is physically consistent. For a broadside point target, the migration around closest approach is approximately

$$R(u) - q \simeq \frac{(u - y)^2}{2q}. \quad (53)$$

For the present aperture and a slant range of about 3.16 km, this migration is smaller than the one-metre range-resolution cell. The carrier phase nevertheless evolves by several cycles, so coherent azimuth focusing remains possible.

13 Several scatterers and coherent superposition

Because the forward model is linear in the complex field, the raw contributions of several scatterers add coherently:

$$s(t, u) = \sum_p s_p(t, u). \quad (54)$$

After focusing, separated scatterers produce translated and complex-weighted copies of the PSF. If two responses overlap, they may interfere constructively or destructively; image magnitudes do not simply add.

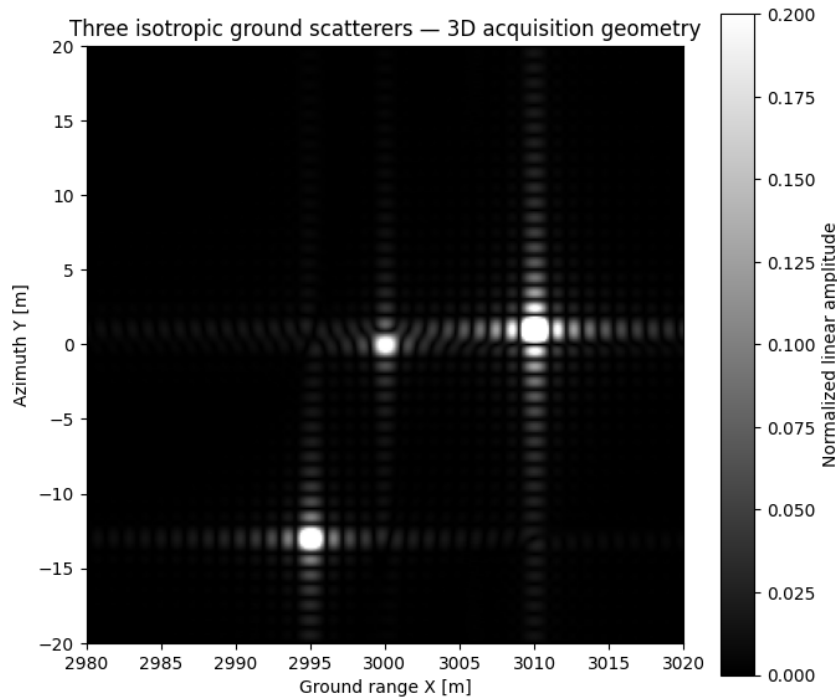


Figure 6: Reconstruction of three point scatterers with different complex reflectivities. The same processing chain is used without modification.

This coherent superposition is the elementary mechanism from which speckle emerges when many unresolved scatterers contribute within one resolution cell.

14 Correspondence with the Python implementation

The central reconstruction sequence is therefore

$$s(t, u) \xrightarrow{\mathcal{F}_t + H_r} S_r(k, u) \xrightarrow{\mathcal{F}_u} S_r(k, k_y) \xrightarrow{H_{2D}} S_c(k, k_y) \xrightarrow{Stolt} F(K_q, k_y) \xrightarrow{\mathcal{F}_{2D}^{-1}} \hat{\rho}_q(q, y) \xrightarrow{q \rightarrow X} \hat{\rho}_g(X, y). \quad (55)$$

15 Assumptions and limits

The compact model intentionally assumes a straight trajectory, constant altitude, flat terrain, exact platform positions, broadside stripmap operation, scalar point scatterers, no antenna pattern, no

Table 2: Direct correspondence between the derivation and the notebook implementation.

Mathematical operation	Notebook implementation
Build physical and discrete axes	<code>SARParams.build_axes()</code>
Exact three-dimensional propagation	<code>synthesize_raw_sar_signal_3d()</code>
Reference waveform	<code>make_fast_time_reference()</code>
Range filtering	<code>range_filter()</code>
Aperture Fourier transform	<code>fft_aperture()</code> or padded variant
Dispersion relation	<code>kq_squared = 4*k**2 - ky**2</code>
Two-dimensional phase compensation	<code>H_2d = exp(1j*phase_filter)</code>
Stolt interpolation	per- k_y calls to <code>interp1d()</code>
Optional spectral apodization	Hann or Hamming windows
Inverse two-dimensional FFT	<code>ifft2_centered()</code>
Slant-to-ground conversion	<code>remap_slant_image_to_ground()</code>
PSF localization and FWHM	<code>analyze_point_spread_function()</code>

thermal noise, no ambiguities, no motion errors, and no propagation disturbances. The stationary-phase derivation also replaces the finite aperture by an effectively infinite integration interval and neglects a slowly varying amplitude factor.

These assumptions are appropriate for validating the coherent geometry and the numerical implementation. They are not sufficient for a calibrated operational processor, for which antenna patterns, noise, motion compensation, calibration, ambiguities, and terrain geometry must be treated explicitly.

16 Conclusion

The SAR problem considered here is the inversion of the raw complex data $s(t, u)$ into a two-dimensional estimate of the ground reflectivity $\hat{\rho}(x, y)$. For a flat ground plane and a straight trajectory, the three-dimensional acquisition geometry can be reduced exactly to a two-dimensional problem in cross-track slant coordinate q and azimuth y . After range filtering, the aperture Fourier transform is evaluated asymptotically by stationary phase. This produces the monostatic dispersion relation

$$k_q = \sqrt{4k^2 - k_y^2},$$

which is the central relation of the Range Migration Algorithm. Phase compensation converts the data into samples of the two-dimensional scene spectrum, but on a nonuniform k_q grid. Stolt interpolation restores a Cartesian spectral grid, allowing a two-dimensional inverse FFT. The resulting complex slant-range image is finally remapped to uniform ground range.

The point-target experiment is indispensable: it simultaneously tests propagation signs, factors of two, spectral axes, FFT shifts, phase compensation, interpolation, and geometrical remapping. More fundamentally, it verifies that the implemented reconstruction operator maps a known raw signal back to the correct coefficient in the reflectivity image.

A Stationary-phase theorem used in the derivation

For completeness, consider

$$J(\lambda) = \int a(u) \exp[j\lambda\psi(u)] du, \quad (56)$$

where λ is large, a varies slowly, and ψ has one non-degenerate stationary point u_s satisfying

$$\psi'(u_s) = 0, \quad \psi''(u_s) \neq 0. \quad (57)$$

Then

$$J(\lambda) \sim a(u_s) \sqrt{\frac{2\pi}{\lambda |\psi''(u_s)|}} \exp \left[j\lambda\psi(u_s) + j\frac{\pi}{4} \text{sgn } \psi''(u_s) \right]. \quad (58)$$

In eq. (32), the large oscillation scale is contained directly in k , and the phase curvature is negative. The resulting factor is therefore $\exp(-j\pi/4)$, as shown in eq. (41).

B Sampling, extent, and resolution

For a Cartesian Fourier pair sampled with N points and spectral increment Δk , the spatial sample spacing is

$$\Delta x = \frac{2\pi}{N\Delta k}, \quad (59)$$

and the unaliased spatial extent is

$$L_x = N\Delta x = \frac{2\pi}{\Delta k}. \quad (60)$$

By contrast, the physical resolution is governed by the total occupied spectral support B_k :

$$\delta x \sim \frac{2\pi}{B_k}. \quad (61)$$

Thus:

- wider spectral support improves resolution;
- finer spectral sampling enlarges the unambiguous image extent;
- zero-padding increases display sampling but does not add spectral support.

These distinctions apply independently in slant range and azimuth.

References

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